Sumant Shivaram Gosi

Embedded Software Engineer
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PROFILE

ISTQB certified software engineer with over 4 years of experience in embedded software development, specialising in C and Python programming, model-based development, software integration and test automation.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer September 2023 – Present
IAV GmbH, Sindelfingen, Germany

- Developed embedded software for electric drive and dual-clutch transmission projects of a German premium OEM using MATLAB/Simulink
- Independently audited Simulink models against MXAM guidelines, classifying and resolving violations
- Performed static code analysis with Lint (MISRA-C) and Polyspace to eliminate runtime risks
- Implemented group and global interfaces and validated OEM signals for semantics, data types and scaling
- Integrated software on multi-core ECUs: flashed software via CANape (XCP) and migrated calibration datasets with vCDMstudio
- Validated post-flash behaviour with CAPL scripts in CANoe, checking task timing, multi-core load and EEPROM operations
- Extended internal development tools with MATLAB scripts, fixed existing bugs and implemented new check rules
- Built a Jenkins CI pipeline in Python that runs Simulink model checks automatically, significantly reducing manual testing effort
- Developed a Python tool that automatically validates ECU HiL test results, cutting manual effort from weeks to days

Embedded AI Intern June 2022 – November 2022
Bosch Global Software Technologies, Renningen, Germany

- Developed a deployment pipeline for DNNs on the STM32 NUCLEO-F767ZI using STM32Cube.AI and Zephyr RTOS on Linux
- Integrated the deployment toolchain into an AutoML-based neural architecture search framework with Azure cloud-to-device profiling
- Measured the inference latency of three DNN architectures on the STM32 NUCLEO-F767ZI: STM32Cube.AI achieved up to 9x lower latency than Zephyr RTOS

Software Engineer August 2019 – March 2021
Robert Bosch, Bengaluru, India

- Analysed software requirements for engine management air path functions in IBM DOORS and supported change request assessment and requirements traceability
- Developed air path functions in ETAS ASCET based on the MSR/AUTOSAR architecture and checked the auto-generated C code against MISRA standards
- Developed and executed unit and functional test cases, including SiL, HiL and calibration measurements (ETAS INCA)
- Performed ASPICE-based spot checks on all deliverables, ensuring process compliance across the V-model development cycle

EDUCATION

Master of Science, Automotive Software Engineering April 2021 – August 2023
Chemnitz University of Technology (TU Chemnitz), Chemnitz, Germany
Grade: 1.7 (German scale, 1.0 = best)
Master's thesis (IAV GmbH): Developed a YOLOv8-based algorithm for detecting forgotten objects in autonomous shuttles and validated it on real-world data (mAP50: 90.9%)

Bachelor of Science, Electronics and Communication Engineering June 2015 – April 2019
SDM College of Science and Technology, Dharwad, India
Grade: 1.7 (German scale)

SKILLS

Programming: C, C++, Python (OOP), MATLAB/Simulink, TargetLink, ASCET, CAPL

Testing: SiL (TPT), HiL (ETAS LABCAR, dSPACE ControlDesk, AutomationDesk), Polyspace, ETAS INCA, PyTest, Robot Framework, Appium, IBM RQM

Processes & Standards: V-Model, AUTOSAR (Classic & Adaptive), ASPICE, MISRA-C, MXAM, Agile, ISO 26262

Vector Toolchain: CANoe (incl. Python scripting), CANape (A2L), vTestStudio, vCDMstudio

Firmware: CMake, FreeRTOS, Zephyr RTOS, Lauterbach Trace32 (SWD/JTAG)

Communication Protocols: CAN, CAN FD, UDS (ISO 14229), XCP, Automotive Ethernet (DoIP, SOME/IP), AUTOSAR NM, UART, SPI, I2C, PCAN

DevOps: Jenkins, GitLab CI, GitHub Actions, Docker

Version Control & Documentation: Git, IBM DOORS, Polarion, SBM, PVCS, Jira, Confluence

Languages: English C1, German B2 (currently attending a C1.2 course)

Driving Licence: Class B

PERSONAL PROJECTS

stm32-freertos-deepdive

- Implemented FreeRTOS concepts on the STM32 F446RE, including multitasking, queues, mutexes, semaphores, software timers, priority inversion and DMA-based non-blocking UART communication with ISR-safe synchronisation

stm32-can-distributed-system

- Built a two-node CAN bus system at 500 kbit/s on the STM32 F446RE with threshold monitoring, a custom ACK protocol and a FreeRTOS task architecture

ml-inference-service

- Developed a production-ready machine learning inference service with FastAPI, Docker containerisation and pytest tests, plus an automated CI/CD pipeline with GitHub Actions for AWS EC2 deployment and Kubernetes manifests for scalable deployment

TRAINING

Python For Automotive **In progress, 2026**

Jacks Technologies AET4 **Completed 2026**

CERTIFICATIONS

ISTQB Certified Tester Foundation Level **July 2024**
Internationally recognised certification in software testing methodology, test planning and quality assurance

telc Deutsch B2 **August 2025**
Official certification of German language proficiency at level B2 of the Common European Framework of Reference for Languages (CEFR)

AWARDS

Deutschlandstipendium (Germany Scholarship) **November 2022 – November 2023**
Chemnitz University of Technology: awarded the Deutschlandstipendium for outstanding academic performance during the Master's programme